

# Anirudh Ravipudi

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## Professional Summary

Data and AI/ML professional with hands-on experience building scalable pipelines, machine learning systems, and NLP applications in financial services. Proficient across the modern data stack, Databricks, Snowflake, dbt, PySpark, and AWS, with a track record of delivering end-to-end workflows that reduce data preparation time and accelerate model development. Experienced in building Power BI dashboards, integrating LLM APIs into production systems, and developing React Native applications with AI-driven features.

## Experience

### AI Cloud Solutions | Data Engineer

Aug 2024 - Present

- Designed and deployed unified batch and real-time data pipelines on Databricks and AWS (S3, EC2, Redshift), including real-time Kafka ingestion pipelines, processing structured and semi-structured transaction data from multiple sources and reducing data preparation time by 30%.
- Built and optimized PySpark jobs on Databricks using Delta Lake for ACID-compliant data storage, implementing partitioning and parallel processing strategies that reduced pipeline latency and improved throughput for downstream analytics and ML workflows.
- Developed modular data transformation layers using dbt on Snowflake, building reusable fact and dimension table models that standardized reporting logic and reduced redundant SQL across analytics teams, and managed data governance and lineage tracking via Databricks Unity Catalog."
- Engineered behavioral and transactional feature pipelines generating signals for predictive models, including transaction velocity, rolling aggregations, and anomaly indicators, reducing model preparation time by 20%.
- Built and maintained interactive dashboards in Power BI connected to Snowflake, enabling business stakeholders to monitor financial KPIs and transaction trends in real time.
- Developed FastAPI endpoints to expose model outputs and pipeline results for real-time consumption by downstream applications, with average prediction retrieval latency under 200ms.
- Containerized pipeline services and ML APIs using Docker and Kubernetes with automated CI/CD via Jenkins and GitHub, enabling scalable deployment and minimizing production errors.

### FoxFire Technologies | Data Analyst Intern

Jan 2021 - Dec 2021

- Applied statistical techniques including hypothesis testing and anomaly detection to validate data quality across reporting pipelines, ensuring reliability of analytical outputs used in business decisions.
- Designed and maintained interactive dashboards in Tableau and Power BI to monitor KPIs in real time, enabling stakeholders to track performance metrics and act faster on emerging trends.

## Projects

### Fraud Detection Data Pipeline & ML System

- Built a scalable PySpark ETL pipeline orchestrated with Airflow to ingest and transform 284,807 real credit card transactions from PostgreSQL, cleaning and writing processed Parquet files to AWS S3 for downstream ML workflows.
- Engineered 5 behavioral features including log-normalized transaction amounts, rolling transaction velocity over a 100-row window, rolling average amount, transaction hour, and a high-value transaction flag to improve model signal quality.
- Trained Random Forest and XGBoost classifiers to detect fraud, achieving 0.98 ROC-AUC while handling severe class imbalance (0.17% fraud rate) using class weighting and scale\_pos\_weight techniques, selecting Random Forest with 83% recall and 77% precision on 56,962 held-out transactions.
- Deployed fraud prediction service via a FastAPI REST endpoint containerized with Docker and managed on Kubernetes, returning fraud labels and probability scores within 200ms to support real-time decisioning.

### LLM Powered Text Analytics & Summarization System

- Built an NLP ingestion pipeline to store and query earnings call transcripts for 5 major companies (AAPL, MSFT, GOOGL, AMZN, NVDA) in PostgreSQL, with metadata filtering by ticker, period, and report type.
- Implemented abstractive summarization using facebook/bart-large-cnn and financial sentiment analysis using ProsusAI/finbert, a domain-specific BERT model fine-tuned on financial text, correctly identifying Apple as the only negative report at 94.88% confidence.
- Extracted key bigrams using YAKE keyword extraction and exposed all results via a FastAPI REST API with 4 endpoints for on-demand analysis, sentiment aggregation, and ticker-based retrieval, with models loaded at startup to minimize per-request latency.

### AI Companion - Mobile Application

- Built a cross-platform mobile and web application using React Native and Expo, integrating the Anthropic Claude API to power emotionally intelligent conversations with a custom persona and adaptive response system.
- Designed a semantic long-term memory architecture using PostgreSQL and pgvector, generating OpenAI text embeddings to enable cosine similarity search across stored user memories, surfacing relevant context in real time during conversations.
- Engineered a multi-signal emotion detection pipeline combining behavioral text analysis, keyword pattern matching, and few-shot classification using 1,038 labeled examples sourced from GoEmotions and EmpatheticDialogues datasets to infer user sentiment without explicit input.
- Implemented a mood-adaptive UI theming system in React Native that transitions color palettes across 12 emotional states based on real-time sentiment inference returned from the backend.
- Developed a Node.js/Express REST API with 8 endpoints handling conversation persistence, semantic memory retrieval, proactive check-in scheduling via node-cron, and third-party integrations including OpenWeatherMap and NewsAPI.
- Configured Supabase with row-level security policies, pgvector extension, and a custom SQL similarity search function, ensuring all user data is privately scoped and never cross-accessible between accounts.

## Technical Skills

- Programming & Scripting:** Python, SQL, R, Java, C/C++, Javascript
- Data Engineering:** Apache Spark, PySpark, Airflow, Kafka, ELT/ETL Pipelines, Snowflake, Databricks, Data Modeling, Data Warehousing, Delta Lake, dbt, PostgreSQL
- Cloud & Infrastructure:** AWS (S3, EC2, IAM, Redshift), Kubernetes, Docker, Azure, Supabase
- APIs & Backend:** FastAPI, REST APIs, Node.js, Express
- Analytics & Machine Learning:** NumPy, pandas, scikit-learn, Feature Engineering, Statistical Methods, Model Evaluation, Supervised Learning, NLP, pgvector, OpenAI Embeddings, Few-Shot Classification
- Visualization & Tools :** Power BI, Tableau, Looker, Git, GitHub, Jenkins, CI/CD

## Education

Florida International University | Master's, Data Science & Artificial Intelligence

Miami, FL